

Proof of Theorem 1

From the definition of the conditional distribution,

$$q_{\mathcal{R}}(\hat{\mathbf{r}} = \mathbf{0} \mid \mathbf{x}_0) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{\|\mathcal{R}(\mathbf{x}_0)\|^2}{2\sigma^2}\right). \quad (1)$$

Taking the logarithm yields:

$$\log q_{\mathcal{R}}(\hat{\mathbf{r}} = \mathbf{0} \mid \mathbf{x}_0) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \|\mathcal{R}(\mathbf{x}_0)\|^2. \quad (2)$$

Substituting this into the objective function gives:

$$\mathcal{L}(\Theta) = \mathbb{E}_{\mathbf{x}_0 \sim p_{\Theta}(\mathbf{x}_0)} \left[-\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \|\mathcal{R}(\mathbf{x}_0)\|^2 \right]. \quad (3)$$

Since the first term is independent of Θ , maximizing $\mathcal{L}(\Theta)$ is equivalent to maximizing the expectation of the second term, which in turn is equivalent to minimizing $\mathbb{E}_{p_{\Theta}} \left[\frac{1}{2\sigma^2} \|\mathcal{R}(\mathbf{x}_0)\|^2 \right]$. This completes the proof of the first equivalence.

For the gradient, we first compute the gradient of the log conditional probability with respect to $\mathbf{x}_0$:

$$\begin{aligned} \nabla_{\mathbf{x}_0} \log q_{\mathcal{R}}(\hat{\mathbf{r}} = \mathbf{0} \mid \mathbf{x}_0) &= \frac{1}{\sigma^2} (\nabla_{\mathbf{x}_0} \mathcal{R}(\mathbf{x}_0))^{\top} (\hat{\mathbf{r}} - \mathcal{R}(\mathbf{x}_0)) \Big|_{\hat{\mathbf{r}}=\mathbf{0}} \\ &= -\frac{1}{\sigma^2} (\nabla_{\mathbf{x}_0} \mathcal{R}(\mathbf{x}_0))^{\top} \mathcal{R}(\mathbf{x}_0). \end{aligned} \quad (4)$$

Under the reparameterization, applying the chain rule yields:

$$\begin{aligned} \nabla_{\Theta} \mathcal{L}(\Theta) &= \mathbb{E}_{\epsilon \sim p(\epsilon)} [\nabla_{\mathbf{x}_0} \log q_{\mathcal{R}}(\hat{\mathbf{r}} = \mathbf{0} \mid \mathbf{x}_0) \nabla_{\Theta} \mathbf{x}_0] \\ &= \mathbb{E}_{\epsilon} \left[-\frac{1}{\sigma^2} (\nabla_{\mathbf{x}_0} \mathcal{R}(\mathbf{x}_0))^{\top} \mathcal{R}(\mathbf{x}_0) \nabla_{\Theta} s_{\Theta}(\epsilon) \right], \end{aligned} \quad (5)$$

which gives the desired gradient expression. If we define the loss function $J(\Theta) = -\mathcal{L}(\Theta)$, then the gradient descent direction becomes $\frac{1}{\sigma^2} \mathbb{E} \left[(\nabla_{\mathbf{x}_0} \mathcal{R}(\mathbf{x}_0))^{\top} \mathcal{R}(\mathbf{x}_0) \nabla_{\Theta} \mathbf{x}_0 \right]$, which adjusts the parameters precisely in the direction that reduces the residual. This completes the proof.

Proof of Theorem 2

Since ∇E is L -Lipschitz continuous, the descent lemma yields

$$E(\mathbf{x}_{\tau_{i-1}}) \leq E(\mathbf{x}_{\tau_i}) + \langle \nabla E(\mathbf{x}_{\tau_i}), \mathbf{x}_{\tau_{i-1}} - \mathbf{x}_{\tau_i} \rangle + \frac{L}{2} \|\mathbf{x}_{\tau_{i-1}} - \mathbf{x}_{\tau_i}\|^2.$$

Let $\mathbf{d}_i = \mathbf{x}_{\tau_{i-1}} - \mathbf{x}_{\tau_i}$. Then

$$E(\mathbf{x}_{\tau_{i-1}}) \leq E(\mathbf{x}_{\tau_i}) + \langle \nabla E(\mathbf{x}_{\tau_i}), \mathbf{d}_i \rangle + \frac{L}{2} \|\mathbf{d}_i\|^2.$$

By the assumed sufficient descent property,

$$\langle \nabla E(\mathbf{x}_{\tau_i}), \mathbf{d}_i \rangle \leq -\mu \|\mathbf{d}_i\|^2.$$

Therefore,

$$E(\mathbf{x}_{\tau_{i-1}}) \leq E(\mathbf{x}_{\tau_i}) - \left(\mu - \frac{L}{2}\right) \|\mathbf{d}_i\|^2 \leq E(\mathbf{x}_{\tau_i}),$$

where the last inequality follows from $\mu > L/2$. This proves the claim.

Convergence of the enhanced sampling trajectory

Let

$$E(\mathbf{x}) = \frac{1}{2\sigma^2} \|\mathcal{R}(\mathbf{x})\|^2$$

be the physics-informed energy function. Consider the deterministic accelerated reverse trajectory

$$\{\mathbf{x}_{\tau_i}\}_{i=0}^S, \quad \tau_S = T, \tau_0 = 0, S \ll T,$$

generated by

$$\mathbf{x}_{\tau_{i-1}} = \sqrt{\bar{\alpha}_{\tau_{i-1}}} \left(\frac{\mathbf{x}_{\tau_i} - \sqrt{1 - \bar{\alpha}_{\tau_i}} \boldsymbol{\epsilon}_{\Theta^*}(\mathbf{x}_{\tau_i}, \tau_i)}{\sqrt{\bar{\alpha}_{\tau_i}}} \right) + \sqrt{1 - \bar{\alpha}_{\tau_{i-1}}} \boldsymbol{\epsilon}_{\Theta^*}(\mathbf{x}_{\tau_i}, \tau_i), \quad i = S, \dots, 1.$$

Define the increment

$$\mathbf{d}_i := \mathbf{x}_{\tau_{i-1}} - \mathbf{x}_{\tau_i}.$$

Assume that:

1. E is continuously differentiable and bounded below;
2. ∇E is L -Lipschitz continuous along the trajectory, i.e.,

$$\|\nabla E(\mathbf{x}) - \nabla E(\mathbf{y})\| \leq L \|\mathbf{x} - \mathbf{y}\|;$$

3. there exists a constant $\mu > \frac{L}{2}$ such that

$$\langle \nabla E(\mathbf{x}_{\tau_i}), \mathbf{d}_i \rangle \leq -\mu \|\mathbf{d}_i\|^2, \quad i = 1, \dots, S.$$

Then the following hold:

$$E(\mathbf{x}_{\tau_{i-1}}) \leq E(\mathbf{x}_{\tau_i}), \quad i = 1, \dots, S, \tag{6}$$

$$E(\mathbf{x}_{\tau_{i-1}}) \leq E(\mathbf{x}_{\tau_i}) - \left(\mu - \frac{L}{2}\right) \|\mathbf{d}_i\|^2, \tag{7}$$

$$\sum_{i=1}^S \|\mathbf{d}_i\|^2 \leq \frac{E(\mathbf{x}_{\tau_S}) - E_{\inf}}{\mu - L/2}, \quad E_{\inf} := \inf_{\mathbf{x}} E(\mathbf{x}). \tag{8}$$

Moreover, if the trajectory is extended indefinitely and there exists a constant $c > 0$ such that

$$\|\mathbf{d}_i\| \geq c \|\nabla E(\mathbf{x}_{\tau_i})\|,$$

then

$$\lim_{i \rightarrow \infty} \|\nabla E(\mathbf{x}_{\tau_i})\| = 0,$$

and every accumulation point of the trajectory is a stationary point of E .

We prove the result in four steps.

Step 1: Descent lemma. Since E is continuously differentiable and ∇E is L -Lipschitz continuous, the descent lemma implies that for any $\mathbf{x}, \mathbf{y}$,

$$E(\mathbf{y}) \leq E(\mathbf{x}) + \langle \nabla E(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|^2.$$

Applying this with $\mathbf{x} = \mathbf{x}_{\tau_i}$ and $\mathbf{y} = \mathbf{x}_{\tau_{i-1}}$, we obtain

$$E(\mathbf{x}_{\tau_{i-1}}) \leq E(\mathbf{x}_{\tau_i}) + \langle \nabla E(\mathbf{x}_{\tau_i}), \mathbf{x}_{\tau_{i-1}} - \mathbf{x}_{\tau_i} \rangle + \frac{L}{2} \|\mathbf{x}_{\tau_{i-1}} - \mathbf{x}_{\tau_i}\|^2.$$

Using the notation $\mathbf{d}_i = \mathbf{x}_{\tau_{i-1}} - \mathbf{x}_{\tau_i}$, this becomes

$$E(\mathbf{x}_{\tau_{i-1}}) \leq E(\mathbf{x}_{\tau_i}) + \langle \nabla E(\mathbf{x}_{\tau_i}), \mathbf{d}_i \rangle + \frac{L}{2} \|\mathbf{d}_i\|^2.$$

Step 2: Monotonic decrease. By the sufficient descent assumption,

$$\langle \nabla E(\mathbf{x}_{\tau_i}), \mathbf{d}_i \rangle \leq -\mu \|\mathbf{d}_i\|^2.$$

Substituting this into the previous inequality yields

$$E(\mathbf{x}_{\tau_{i-1}}) \leq E(\mathbf{x}_{\tau_i}) - \mu \|\mathbf{d}_i\|^2 + \frac{L}{2} \|\mathbf{d}_i\|^2 = E(\mathbf{x}_{\tau_i}) - \left(\mu - \frac{L}{2}\right) \|\mathbf{d}_i\|^2.$$

Since $\mu > \frac{L}{2}$, it follows immediately that

$$E(\mathbf{x}_{\tau_{i-1}}) \leq E(\mathbf{x}_{\tau_i}), \quad i = 1, \dots, S.$$

This proves the monotonic energy decrease and the quantitative decrease bound.

Step 3: Finite-length property. Summing

$$E(\mathbf{x}_{\tau_{i-1}}) \leq E(\mathbf{x}_{\tau_i}) - \left(\mu - \frac{L}{2}\right) \|\mathbf{d}_i\|^2$$

over $i = 1, \dots, S$, we obtain

$$\sum_{i=1}^S \left(\mu - \frac{L}{2}\right) \|\mathbf{d}_i\|^2 \leq E(\mathbf{x}_{\tau_S}) - E(\mathbf{x}_{\tau_0}).$$

Because E is bounded below,

$$E(\mathbf{x}_{\tau_0}) \geq E_{\inf}.$$

Hence,

$$\sum_{i=1}^S \|\mathbf{d}_i\|^2 \leq \frac{E(\mathbf{x}_{\tau_S}) - E_{\inf}}{\mu - L/2}.$$

Step 4: Convergence to stationary points. Now suppose the trajectory is extended to infinitely many steps and that there exists $c > 0$ such that

$$\|\mathbf{d}_i\| \geq c \|\nabla E(\mathbf{x}_{\tau_i})\|.$$

Using the decrease estimate from Step 2,

$$E(\mathbf{x}_{\tau_{i-1}}) \leq E(\mathbf{x}_{\tau_i}) - \left(\mu - \frac{L}{2}\right) \|\mathbf{d}_i\|^2,$$

and the gradient-related condition, we get

$$E(\mathbf{x}_{\tau_{i-1}}) \leq E(\mathbf{x}_{\tau_i}) - \left(\mu - \frac{L}{2}\right) c^2 \|\nabla E(\mathbf{x}_{\tau_i})\|^2.$$

Summing this inequality over i shows that

$$\sum_{i=1}^{\infty} \|\nabla E(\mathbf{x}_{\tau_i})\|^2 < \infty.$$

Therefore,

$$\|\nabla E(\mathbf{x}_{\tau_i})\| \rightarrow 0 \quad \text{as } i \rightarrow \infty.$$

Let $\mathbf{x}^*$ be any accumulation point of the sequence $\{\mathbf{x}_{\tau_i}\}$. By continuity of ∇E ,

$$\nabla E(\mathbf{x}^*) = 0.$$

Thus every accumulation point is a stationary point of E .